

Aarush Narayanan

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Education

B.S. Data Science
University of California San Diego

June 2028

Experience

Software Engineering Intern, The Kaizen Academy – Remote January 2026 – Present

- Architected and deployed a scalable Supabase (Postgres) backend for a production Learning Management System, designing normalized schemas and relational integrity across users, classes, enrollments, assignments, grades, and attendance; implemented SQL-backed API routes integrated into a live Next.js application
- Led end-to-end development of a data-driven Teacher Portal enabling attendance tracking, grading, report card generation, class creation, and student enrollment management; built and validated full CRUD workflows, optimized database queries, and ensured real-time synchronization across role-based dashboards (teachers, students, parents).

Technical Associate, Triton Consulting Group October 2025 – Present

- Rebuilt Triton Consulting Group's marketing site using Next.js and Vercel, implementing responsive layouts and interactive UI components with React and CSS, and restructuring site navigation to improve content accessibility for 300+ quarterly applicants

Software Engineer Intern, SmartSolar – Ho Chi Minn City, Vietnam (Remote) June 2024 – August 2024

- Integrated Google Maps and Leaflet.js APIs to build an MVP solar quote tool in a Webflow-based frontend, enabling users to enter their address and draw installation areas, helping advance solar adoption in Vietnam
- Designed full-stack geospatial quote-calculation tool system with slug-based data storage and optimized algorithms, cutting quote generation time by 40%

Student Researcher, Aspiring Student Directed Research Program – Fremont, CA Dec 2022 – Dec 2024

- Researched correlation between college campus environment and retention rates using over 3000 google street view images by developing K-means cluster algorithm using scikit-learn to extract color palettes
- Developed QCNN to analyze images of pigmented skin lesions and diagnose the lesion type with a 92% accuracy
- Applied stochastic time-series modeling (ARIMA & Prophet), feature engineering, and data preprocessing to forecast ozone pollution in Long Beach, CA with 0.1% error, gaining expertise in predictive analytics and environmental data modeling

Projects

CRYSTL | Python, OpenCV, YOLO, Tensorflow, PyTorch December 2024 - May 2025

- Developed a computer vision model utilizing clip-on camera system to detect alcohol bottles in real time to support recovering individuals by providing early-warning relapse prevention
- Developed and trained custom object-detection pipeline using YOLO architectures in PyTorch/TensorFlow, utilizing dataset I built of 2000+ images of alcohol bottles and applying hyperparameter tuning
- Collected and analyzed data from over 100 individuals to finetune model

VeriScope | React, FastAPI, RoBERTa, AWS, PostgreSQL, Docker December 2025 - Present

- Architected and shipped a production-ready, containerized fake news detection platform using React (AWS S3), FastAPI (Docker on AWS ECS Fargate), RoBERTa-based NLP models, and PostgreSQL (AWS RDS), enabling scalable, low-latency inference across 50,000+ real and fake news articles.
- Improved model performance from 33% to 77% accuracy by designing a calibrated evaluation pipeline, running simulations on recent real-world news data, and applying temperature and Platt scaling to stabilize predictions.
- Reduced manual article analysis time by 93% by delivering real-time inference (30 seconds or less per article vs. 7 minutes of human reading), while implementing user history tracking, URL-based ingestion, and feedback labeling to support continuous model refinement.

Skills

Languages: Python, JavaScript, Java, SQL

Technologies: Next.js, React, FastAPI, Supabase, PostgreSQL, Microsoft SQL Server, AWS (S3, ECS Fargate, RDS), Docker, PyTorch, TensorFlow, scikit-learn, pandas, RoBERTa, YOLO, ARIMA, Prophet, BeautifulSoup, newspaper4k, trafilatura